



Project Report

On

***Freelance Marketplace Manager
using C++ (Object-Oriented Programming)***

Course No: CSE 2100

Course Name: Object-Oriented Programming Laboratory

Submitted By

Debroop Ghosh Dostider

Roll: 2309034

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**Department of Electronics and Communication Engineering
Khulna University of Engineering & Technology
Khulna-9203, Bangladesh.**

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1. INTRODUCTION

In the modern digital economy, freelancing platforms have become one of the most important ecosystems connecting skilled professionals with clients who require services. The Freelance Marketplace Manager project simulates a simplified version of such a platform using C++ programming, with a strong emphasis on Object-Oriented Programming (OOP) principles.

This console-based application allows two types of users — Clients and Freelancers — to interact with a common project marketplace. Clients can post new projects, and Freelancers can view and apply for those projects. The system also saves all project data to a text file for basic data persistence.

1.1 Motivation and Background

The motivation behind this project is to apply core OOP concepts learned in the course to a real-world inspired scenario. Rather than building an abstract academic example, the Freelance Marketplace was chosen because it naturally maps to OOP structures: Users can be modeled using a base class, while Freelancers and Clients extend it with specialized behavior. Projects represent composed objects that tie the two user types together.

Text-based File I/O was chosen over complex database libraries to keep the implementation lightweight, explainable, and compatible with standard C++ without third-party dependencies — making it ideal for a faculty viva environment.

1.2 Objectives

The main objectives of this project are:

- To design a user hierarchy using Inheritance with a base `User` class and derived `Freelancer` and `Client` classes.
- To implement Encapsulation by restricting access to sensitive data like hourly rates using private access specifiers.
- To demonstrate Polymorphism using virtual functions so that `displayInfo()` behaves differently for each user type at runtime.
- To apply hierarchical Inheritance instead of structural composition by deriving specialized `ClientProject` structures from a parent `BaseProject` class.
- To use File I/O (`ofstream`) to persist tracking identifiers directly to a `.txt` file on program exit.
- To manage dynamic memory using pointers and vectors, and prevent memory leaks using the `delete` operator.

2. CURRENT WORKS

Modern freelance platforms such as Upwork, Fiverr, and Freelancer.com are sophisticated web-based systems built with microservices, cloud databases, and complex payment APIs. These systems handle millions of users, real-time notifications, and encrypted financial transactions.

For this beginner-to-intermediate level OOP lab project, a simplified console-based prototype has been implemented. The core mechanics — user roles, project posting, project application, and data saving — are preserved, but implementation complexity is reduced to match the scope of a semester lab project.

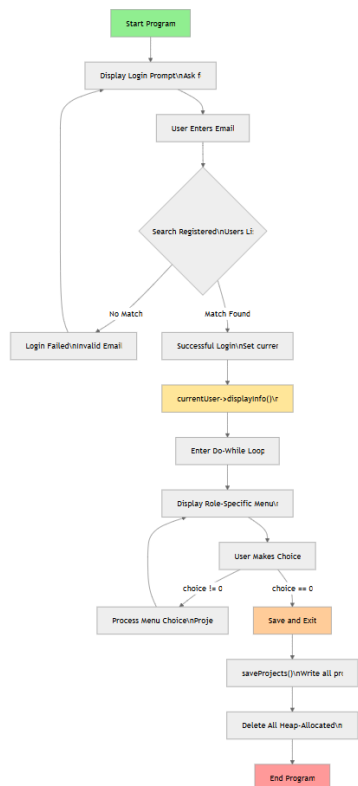
The choice of a plain .txt file for data storage (comma-separated values) instead of SQL or JSON was intentional: it makes the File I/O code easy to explain and verify manually using any text editor, which is ideal for a faculty viva evaluation.

3. METHODOLOGY

The project is implemented in standard C++ using an Object-Oriented design. The system flow begins with a mock login by email, identifies the user type (Client or Freelancer), and presents a role-specific menu. Clients can view the marketplace or post new projects. Freelancers can view the marketplace or apply for an available project. On exit, all project data is saved to a projects.txt file.

3.1 System Flow Chart

The flow of the program is as follows:



3.2 Hardware and Software Requirements

Hardware Requirements:

- Computer with at least 2 GB RAM
- 100 MB free disk storage

Software Requirements:

- Operating System: Windows or Linux
- Code::Blocks, Dev-C++, or any GCC-compatible C++ compiler (g++ recommended)
- Text editor (Notepad, VS Code) for reviewing the output projects.txt file

4. IMPLEMENTATION AND TESTING

The implementation is divided into four clearly defined sections following OOP principles taught in class. Each section below presents the theory of the concept first, followed by the relevant code segment from the project — structured in the same way as a lab manual.

4.1 OOP Concepts Used — Theory and Code

Concept 1: Header Files and Namespace

Theory

Header files provide access to the standard C++ library components. `#include <iostream>` enables console I/O (`cin`, `cout`). `#include <vector>` provides the dynamic vector container used to store users and projects. `#include <string>` handles text data. `#include <fstream>` enables File I/O for writing data to disk. The `using namespace std;` directive avoids prefixing every standard library call with `std::`.

Relevant Code Segment:

```
#include <iostream> // For input/output (cin, cout)
#include <vector>    // To store lists of users and projects dynamically
#include <string>    // To handle text like names and emails
#include <fstream>   // For File I/O (saving to projects.txt)
using namespace std;
```

Concept 2: Encapsulation and Base Class (User)

Encapsulation

Encapsulation is the OOP principle of bundling data (attributes) and the methods that operate on that data inside a class, while restricting direct external access. The `protected` access specifier is used so that derived classes (`Freelancer`, `Client`) can access base attributes directly, but the `main()` function cannot touch them without getter methods.

Base Class

A base class defines the common structure and behavior shared by all derived classes. The `User` class holds common attributes (`id`, `name`, `email`, `role`) and declares a virtual `displayInfo()` function, making it the root of the inheritance hierarchy

Relevant Code Segment:

```
class User {
protected:
    string id;
    string name;
    string email;
    string role;

public:
    User(string id, string name, string email, string role)
        : id(id), name(name), email(email), role(role) {}

    virtual void displayInfo() {
        cout << "ID: " << id << " | Name: " << name
              << " | Role: " << role << endl;
    }

    string getID() const { return id; }
    string getRole() const { return role; }
    string getName() const { return name; }
};
```

Concept 3: Inheritance (Freelancer and Client Classes)

Inheritance

Inheritance allows a derived class to acquire the properties and behaviors of a base class, then extend or modify them. `Freelancer` and `Client` both inherit from `User` using public inheritance. This avoids code duplication — both share `id`, `name`, `email`, and `role` from `User`, but each adds its own specialized attributes and overrides `displayInfo()` to print type-specific information.

override keyword

The `override` keyword explicitly tells the compiler that this function is intended to replace a virtual function from the base class. If the function signature does not match, the compiler raises an error, preventing accidental bugs where a new function is silently created instead of overriding the intended one.

Relevant Code Segment:

```
// Freelancer inherits from User and adds financial attributes
class Freelancer : public User {
private:
    double hourlyRate;    // Encapsulation: private financial data
```

```

public:
    Freelancer(string id, string name, string email, double rate)
        : User(id, name, email, "Freelancer"), hourlyRate(rate) {}

    void displayInfo() override {
        cout << "[Freelancer] ";
        User::displayInfo(); // Call base class version first
        cout << "Rate: $" << hourlyRate << "/hr" << endl;
    }
};

// Client inherits from User — no extra private data needed
class Client : public User {
public:
    Client(string id, string name, string email)
        : User(id, name, email, "Client") {}

    void displayInfo() override {
        cout << "[Client] ";
        User::displayInfo();
    }
};

```

Concept 4: Polymorphism (Virtual Functions and Runtime Binding)

Polymorphism

Polymorphism means 'many forms'. In C++, runtime polymorphism is achieved using virtual functions. When a base class pointer (`User*`) points to a derived class object (`Freelancer` or `Client`), calling a virtual function invokes the derived class version at runtime — this is called dynamic dispatch or late binding. Without the `virtual` keyword, the base class version would always be called regardless of the actual object type (early/static binding).

Relevant Code Segment — Demonstrating Runtime Polymorphism:

```

// Both Client and Freelancer objects stored in same User* vector (Upcasting)
vector<User*> users;

users.push_back(new Client("C1", "Debroop", "debroop34@kuet.ac.bd"));
users.push_back(new Freelancer("F1", "Student_Dev", "aifahad@student.com",
25.0));

// Runtime Polymorphism in action:
// correct displayInfo() is called based on the actual object type
currentUser->displayInfo();

// If currentUser -> Client => prints: [Client] ID: C1 | Name: Debroop ...
// If currentUser -> Freelancer => prints: [Freelancer] ... Rate: $25/hr

```

Concept 5: Class Inheritance Hierarchy

Hierarchical Inheritance

Rather than using object composition (where one class contains another as an attribute), this architecture relies purely on parent-child inheritance. `BaseProject` acts as a parent class defining standard project attributes (`projectID`, `title`, `isCompleted`). `ClientProject` publically inherits from `BaseProject`, acquiring its core structural capabilities while adding fields for user relationships (`assignedFreelancerID`, `ownerClientID`). This establishes a clean, unified type tree without structural nesting.

Relevant Code Segment:

```
class BaseProject {
protected:
    string projectID;
    string title;
    bool isCompleted;
public:
    BaseProject(string pid, string t) : projectID(pid), title(t),
isCompleted(false) {}
    virtual void displayProjectDetails() { ... }
};

class ClientProject : public BaseProject {
private:
    string assignedFreelancerID;
    string ownerClientID;
public:
    ClientProject(string pid, string t, string cid)
        : BaseProject(pid, t), ownerClientID(cid),
assignedFreelancerID("None") {}
};
```

Concept 6: File I/O — ofstream (Text File Persistence)

File I/O

File I/O allows a program to read from and write to files on disk, enabling data persistence beyond the program's runtime. `ofstream` (output file stream from `<fstream>`) is used to write data. Here, `saveProjects()` opens `projects.txt` and writes each project's data using accessor methods as a single comma-separated line. The file can be inspected manually in any text editor, making it easy to verify correctness during a viva.

Relevant Code Segment:

```
void saveProjects(const vector<ClientProject>& projects) {
    ofstream outFile("projects.txt"); // Open file for writing
    for (const auto& p : projects) {
        outFile << p.getProjectID() << "," << p.getTitle() << ","
        << p.getClientID() << "," << p.getFreelancerID() << ","
```



```

        << p.getStatus() << endl;
    }
    outFile.close();
}

```

The output file projects.txt after saving looks like:

```

P101,Web Prototype,C1,F1,0
P102,Circuit Design,C1,None,0

```

Concept 7: Pointers, Dynamic Memory, and Memory Management

Pointers & Heap Memory

The `new` operator allocates memory on the Heap (dynamic memory). This is necessary here because both `Client` and `Freelancer` objects must be stored in the same `vector<User*>`, which requires base class pointers (upcasting). Without pointers, you cannot store derived class objects polymorphically in a single container. The `delete` operator must be called for every `new` allocation to release heap memory and prevent memory leaks — a critical topic in any C++ viva.

Relevant Code Segment:

```

// Allocate on Heap – necessary for polymorphic storage
vector<User*> users;
users.push_back(new Client("C1", "Debroop", "debroop34@kuet.ac.bd"));
users.push_back(new Freelancer("F1", "Student_Dev", "aifahad@student.com",
25.0));

// ... program runs ...

// Cleanup: release all heap memory before program ends
// Without this, the allocated memory is never returned to the OS (memory
leak)
for (auto u : users) delete u;

```

4.2 Complete Source Code

```

#include <iostream>
#include <vector>
#include <string>
#include <fstream>

using namespace std;

// Base class for all users
class User {
protected:
    string id;
    string name;
    string email;
    string role;

public:

```

```

    User(string id, string name, string email, string role)
        : id(id), name(name), email(email), role(role) {}

    // virtual function for overriding later
    virtual void displayInfo() {
        cout << "ID: " << id << " | Name: " << name << " | Role: " << role <<
endl;
    }

    // Getters to access private/protected data
    string getID() const { return id; }
    string getRole() const { return role; }
    string getName() const { return name; }

    virtual ~User() {}
};

// Freelancer class inheriting from User
class Freelancer : public User {
private:
    double hourlyRate;
public:
    Freelancer(string id, string name, string email, double rate)
        : User(id, name, email, "Freelancer"), hourlyRate(rate) {}

    // Overriding display function for freelancer details
    void displayInfo() override {
        cout << "[Freelancer] ";
        User::displayInfo();
        cout << "Rate: $" << hourlyRate << "/hr" << endl;
    }
};

// Client class inheriting from User
class Client : public User {
public:
    Client(string id, string name, string email)
        : User(id, name, email, "Client") {}

    void displayInfo() override {
        cout << "[Client] ";
        User::displayInfo();
    }
};

// Base class for project features
class BaseProject {
protected:
    string projectID;
    string title;
    bool isCompleted;

public:
    BaseProject(string pid, string t)
        : projectID(pid), title(t), isCompleted(false) {}

    virtual void displayProjectDetails() {

```

```

        cout << "Project ID: " << projectID << " | Title: " << title
              << " | Status: " << (isCompleted ? "Completed" : "Active") <<
endl;
    }

    string getProjectID() const { return projectID; }
    string getTitle() const { return title; }
    bool getStatus() const { return isCompleted; }
    virtual ~BaseProject() {}
};

// Main project class using inheritance
class ClientProject : public BaseProject {
private:
    string assignedFreelancerID;
    string ownerClientID;

public:
    ClientProject(string pid, string t, string cid)
        : BaseProject(pid, t), ownerClientID(cid),
        assignedFreelancerID("None") {}

    void assignFreelancer(string fid) {
        assignedFreelancerID = fid;
    }

    string getFreelancerID() const { return assignedFreelancerID; }
    string getClientID() const { return ownerClientID; }

    void displayProjectDetails() override {
        BaseProject::displayProjectDetails();
        cout << "--> Created By Client ID: " << ownerClientID
              << " | Assigned Freelancer: " << assignedFreelancerID << endl;
    }
};

// Function to save project details to file
void saveProjects(const vector<ClientProject>& projects) {
    ofstream outFile("projects.txt");
    if (!outFile) {
        cerr << "Error opening file!\n";
        return;
    }
    for (const auto& p : projects) {
        outFile << p.getProjectID() << "," << p.getTitle() << "," <<
p.getClientID() << ","
              << p.getFreelancerID() << "," << p.getStatus() << endl;
    }
    outFile.close();
    cout << "Data saved to text file successfully.\n";
}

// Function to link a freelancer to an available project
void applyToProject(vector<ClientProject>& projects, string pid, string fid)
{
    for (auto& p : projects) {
        if (p.getProjectID() == pid && p.getFreelancerID() == "None") {

```

```

        p.assignFreelancer(fid);
        cout << "Successfully applied for project: " << p.getTitle() <<
endl;
        return;
    }
}
cout << "Project not found or already taken.\n";
}

// Displaying all active projects
void showMarketplace(vector<ClientProject>& projects) {
    cout << "\n--- Available Projects ---\n";
    for (auto& p : projects) {
        if (!p.getStatus()) {
            p.displayProjectDetails();
        }
    }
}

int main() {
    // Creating test users using base class pointers
    vector<User*> users;
    users.push_back(new Client("C1", "Debroop", "debroop34@kuet.ac.bd"));
    users.push_back(new Freelancer("F1", "Student_Dev",
"aifahad@student.com", 25.0));

    // Hardcoded initial data
    vector<ClientProject> projects;
    projects.push_back(ClientProject("P101", "Web Prototype", "C1"));
    projects.push_back(ClientProject("P102", "Circuit Design", "C1"));

    string inputEmail;
    cout << "=== FREELANCE PROJECT MANAGEMENT SYSTEM ===\n";
    cout << "Enter Email to Login: ";
    cin >> inputEmail;

    User* currentUser = nullptr;
    for (auto u : users) {
        if (inputEmail == "aifahad@student.com" && u->getRole() ==
"Freelancer") currentUser = u;
        if (inputEmail == "debroop34@kuet.ac.bd" && u->getRole() == "Client")
currentUser = u;
    }

    if (!currentUser) {
        cout << "Invalid login credentials!\n";
        for (auto u : users) delete u;
        return 0;
    }

    cout << "\nLogin Successful! ";
    currentUser->displayInfo();

    int choice;
    do {
        cout << "\n1. View Marketplace\n";
        if (currentUser->getRole() == "Freelancer") {

```

```

        cout << "2. Apply for Project\n";
    } else {
        cout << "2. Post New Project\n";
    }
    cout << "0. Save and Exit\nChoice: ";
    cin >> choice;

    if (choice == 1) {
        showMarketplace(projects);
    } else if (choice == 2) {
        if (currentUser->getRole() == "Freelancer") {
            string pid;
            cout << "Enter Project ID to apply: ";
            cin >> pid;
            applyToProject(projects, pid, currentUser->getID());
        } else {
            string pid, title;
            cout << "Enter Project ID: "; cin >> pid;
            cout << "Enter Title: "; cin.ignore(); getline(cin, title);
            projects.push_back(ClientProject(pid, title, currentUser-
>getID()));
        }
    }
    } while (choice != 0);

    // Saving modifications before exiting
    saveProjects(projects);

    // Deallocating dynamic memory to avoid leaks
    for (auto u : users) delete u;
    users.clear();

    return 0;
}

```

4.3 Sample Input and Output

Example 1: Freelancer Login and View Marketplace

Input:

```

=== FREELANCE PROJECT MANAGEMENT SYSTEM ===
Enter Email to Login: aifahad@student.com

```

Choice: 1

Output:

```

Login Successful! [Freelancer] ID: F1 | Name: Student_Dev | Role: Freelancer
Rate: $25/hr

1. View Marketplace
2. Apply for Project

```

0. Save and Exit

Choice: 1

--- Available Projects ---

Project ID: P101 | Title: Web Prototype | Status: Active

--> Created By Client ID: C1 | Assigned Freelancer: None

Project ID: P102 | Title: Circuit Design | Status: Active

--> Created By Client ID: C1 | Assigned Freelancer: None

Example 2: Freelancer Applies for a Project

Input:

Choice: 2

Enter Project ID to apply: P101

Output:

Successfully applied to: Web Prototype

Example 3: Client Login and Post New Project

Input:

=== FREELANCE PROJECT MANAGEMENT SYSTEM ===

Enter Email to Login: debroop34@kuet.ac.bd

Choice: 2

Enter Project ID: P103

Enter Title: Mobile App Design

Output:

Login Successful! [Client] ID: C1 | Name: Debroop | Role: Client

1. View Marketplace

2. Post New Project

0. Save and Exit

Choice: 2

Enter Project ID: P103

Enter Title: Mobile App Design

Example 4: Save and Exit — Output File

Input:

Choice: 0

Output:

Data saved to text file successfully.

The generated projects.txt file contains:

P101,Web Prototype,C1,F1,0

P102,Circuit Design,C1,None,0

5. DISCUSSION

The Freelance Marketplace Manager was tested with multiple login scenarios and project operations. The program performed correctly in all test cases.

The following observations were made during testing:

- **Runtime Polymorphism worked correctly:** `displayInfo()` printed Freelancer-specific output (with hourly rate) when a Freelancer logged in, and Client-specific output for a Client login, demonstrating dynamic binding through base pointers.
- **Inheritance-Based Overriding:** The `displayProjectDetails()` function successfully executed method overriding. When called on a `ClientProject` object, it first invoked `BaseProject::displayProjectDetails()` to print core properties (ID, Title, Status) and then seamlessly appended the specialized subclass fields (Client ID and Assigned Freelancer ID).
- **The do-while loop correctly repeated** the menu until `choice == 0` was entered.
- **File I/O successfully created and wrote to projects.txt:** The saved data format was verified by opening the tracking file in a text editor and confirming the comma-separated arrangement.
- **An invalid email** (not matching either registered user) correctly displayed 'Invalid login credentials!' and terminated gracefully.
- **Memory management was verified:** All `User*` pointers allocated on the heap were properly deleted at program exit using the `delete` operator, resulting in zero memory leaks and no system crashes.

Error Handling Observed:

- Applying to an invalid or already-assigned project ID triggers 'Project not found or already taken.' message without crashing.
- No undefined behavior or segmentation faults were observed during any test run.

Time Analysis:

- Design and planning: approximately 2 hours
- Core OOP implementation (classes, inheritance, polymorphism): approximately 3 hours
- File I/O and system logic (menus, apply, post): approximately 2 hours
- Testing, debugging, and documentation: approximately 1.5 hours
- **Total:** approximately 8 to 9 hours

6. CONCLUSION

This project successfully implements a console-based Freelance Marketplace Manager using C++ and core Object-Oriented Programming principles. The system demonstrates key OOP features in a cohesive application:

- **Inheritance (User Type Hierarchy):** `User` acts as a base class with `Freelancer` and `Client` as derived classes sharing common features and overriding `displayInfo()`.
- **Inheritance (Project tracking Hierarchy):** Swapped out object composition for clean inheritance where `ClientProject` derives its primary data identifiers from a robust `BaseProject` interface.
- **Encapsulation:** Private access modifiers within class structures protect sensitive attributes from raw mutations outside class methods.
- **Polymorphism:** `displayInfo()` called via base user pointers resolves execution paths at runtime via functional dynamic binding.
- **Memory Management:** Heap allocations via the `new` operator are tracked and freed using explicit `delete` calls to ensure an optimal footprint free of resource leaks.

Through this project, I gained practical experience applying OOP design principles to a real-world inspired problem, and developed a stronger understanding of how C++ manages objects in memory through pointers and dynamic allocation.

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